



Technical Service Bulletin

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Intake Valve Chordal Malfunctions

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX15 CM570
- ISX15 CM870
- ISX15 CM871
- ISX15 CM871 E
- ISX15 CM2250
- ISX15 CM2250 SN
- ISX15 CM2350 X101
- QSX15 CM570

Issue

Under severe operating conditions, acidic corrosion and thermal stresses induced in the intake valve can initiate fatigue cracking which, over time, can propagate until a portion of the intake valve separates from the valve head and enters the cylinder. This type of valve malfunction is commonly known as a chordal malfunction. Once the valve has malfunctioned, performance of the engine is compromised and progressive damage to the piston, cylinder liner, cylinder head and turbocharger can occur. Typical complaints following intake valve chordal malfunctions include but are not limited to check engine light, low engine power, cylinder misfire, "popping" noise from the air intake manifold, excessive engine noise, and engine shut down and no start conditions.

Verification

⚠ CAUTION ⚠

Reference Prevention of Turbocharger Damage After Engine Mechanical Issue, Bulletin 4326040, to prevent subsequent engine damage from occurring following repair of a valve malfunction.

- Intake valve chordal malfunctions are **only** one possible cause of the symptoms discussed in this document. The repair facility has the responsibility to follow normal troubleshooting procedures and determine root cause.
- Chordal malfunction of the intake valve can **only** be reliably determined by borescope inspection or removal of the cylinder head.
- A chordal malfunction of an intake valve is characterized by a piece or pieces of valve head missing from an intake valve. Reference Figure 1 and Figure 2 below.
- Visual inspection of turbocharger and exhaust plumbing may reveal the presence of metal debris or, in instances of turbocharger malfunctions, foreign object damage may be observed.
- Fault Codes that may be present following chordal malfunction of an intake valve include:
 - Fault Code 1898 - VGT Actuator Controller - Out of Calibration
 - Fault Code 2387 - VGT Actuator Driver Circuit (Motor) - Mechanical System Not Responding or Out of Adjustment
 - Fault Code 1654, 1655, 1656, 1657, 1658, 1659 – Engine Misfire – Condition Exists



Figure 1, Broken Intake Valve in Cylinder Head.



Figure 2, Broken Intake Valve Showing Radial/Chordal Fracture.

Resolution

- A new intake valve, Part Number 3685996, has been released for assembly in all ISX and QSX cylinder heads.
- To address malfunctions specific to dual cam cylinder heads, a new intake valve insert, Part Number 4393425, has been released for assembly in dual cam ISX and QSX cylinder heads **only**.
- Both the new intake valve and insert are manufactured from premium materials and improve robustness against acidic corrosion and thermal fatigue that are primary contributors to chordal malfunctions.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Cylinder Head and Intake Valve Part Numbers

Part Description	Obsolete Part Number	New Part Number
Cylinder Head, Single Cam	3688761	4386009
Cylinder Head, Dual Cam	4386011	5413784
Cylinder Head, Dual Cam	4331387	5413782
Intake Valve	4965868	3685996
Intake Valve Insert	-	4393425

Part Structure

- The new intake valve is structured into all ISX and QSX cylinder heads.

- The new intake valve seat is structured into dual cam ISX and QSX cylinder heads **only**.

Part Inventory Action

Use existing part number inventory before using new part number.

Production Status

Implemented for production. See Table 2.

Table 2, Production Information				
Phase	Part Number	ESN First	Build Date*	Plant
Introduced Intake Valve	4386009	79832078	18 Apr 2015	Jamestown Engine Plant
	4386011 (Obsolete)	79844115	5 May 2015	
	4331387 (Obsolete)	79764260	6 Aug 2014	
Introduced Valve Seat	5413782**	79939628	1 Sep 2016	
	5413784***	79941325	15 Sep 2016	
*Engine build date can be found on the engine dataplate. **Replaced obsoleted 4331387 ***Replaced obsoleted 4386011 (the head designed for Industrial Tier 3 applications and one particular ISX02 option				

Document History

Date	Details
2015-7-29	Module Created
2016-9-6	Updated resolution, Table 1, and Table 2.

Last Modified: 05-Dec-2016
